Installing Dependencies

```
!pip install langchain langchain_community pypdf
!pip install chromadb
!pip install sentence_transformers
!pip install langchain-ollama
```

```
Requirement already satisfied: websockets>=15.0 in /usr/local/lib/python3.13/dist-packages (from langsmith<1.0.0,>=0.14.0->langchain-core<2.0.0,>=1.2.21
```

```
Requirement already satisfied: websockets>=13.0 in /usr/local/lib/python3.13/dist-packages (from langsmith<1.0.0,>=0.14.9->langchain-core<2.0.0,>=1.2.21
Requirement already satisfied: xxhash>=3.0.0 in /usr/local/lib/python3.13/dist-packages (from langsmith<1.0.0,>=0.14.9->langchain-core<2.0.0,>=1.2.21->l
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.13/dist-packages (from langsmith<1.0.0,>=0.14.9->langchain-core<2.0.0,>=1.2.2
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.13/dist-packages (from pydantic<3.0.0,>=2.7.4->langchain-core<2.0.0,>=1.
Requirement already satisfied: pydantic-core==2.50.1 in /usr/local/lib/python3.13/dist-packages (from pydantic<3.0.0,>=2.7.4->langchain-core<2.0.0,>=1.2
Requirement already satisfied: typing-inspection>=0.4.4 in /usr/local/lib/python3.13/dist-packages (from pydantic<3.0.0,>=2.7.4->langchain-core<2.0.0,>=
Requirement already satisfied: httpcore2==2.13.1 in /usr/local/lib/python3.13/dist-packages (from httpx2<3,>=2->langsmith<1.0.0,>=0.14.9->langchain-core
Requirement already satisfied: truststore>=0.10 in /usr/local/lib/python3.13/dist-packages (from httpx2<3,>=2->langsmith<1.0.0,>=0.14.9->langchain-core<
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests>=2.0.0->langsmith<1.0.0,>=0.14.9->lang
Requirement already satisfied: urllib3<3,>=1.26 in /usr/local/lib/python3.13/dist-packages (from requests>=2.0.0->langsmith<1.0.0,>=0.14.9->langchain-co
Downloading langchain_ollama-1.1.0-py3-none-any.whl (31 kB)
Downloading ollama-0.6.3-py3-none-any.whl (16 kB)
Installing collected packages: ollama, langchain-ollama
Successfully installed langchain-ollama-1.1.0 ollama-0.6.3
```

Ollama Install and setup

```
# Installing Ollama (Linux)
!sudo apt-get install -y zstd
!curl -fsSL https://ollama.com/install.sh | sh

# Setting up Ollama for google colab
# https://colab.research.google.com/github/5aharsh/collama/blob/main/Ollama_Setup.ipynb#scrollTo=Jh5CBAFxBYAC
import threading
import subprocess
import time

def run_ollama_serve():
  subprocess.Popen(["ollama", "serve"])

thread = threading.Thread(target=run_ollama_serve)
thread.start()
time.sleep(5)

# Pulling embedding model
!ollama pull mxbai-embed-large

# Pulling LLMs
!ollama pull gemma3:4b
!ollama pull gemma3:12b
!ollama pull phi4-mini
!ollama pull qwen3:8b
#!ollama pull mistral:7b-instruct
#!ollama pull deepseek-r1:8b

# Checking models
!ollama list
```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  zstd
0 upgraded, 1 newly installed, 0 to remove and 33 not upgraded.
Need to get 644 kB of archives.
After this operation, 1,845 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 zstd amd64 1.5.5+dfsg2-2build1.1 [644 kB]
Fetched 644 kB in 1s (533 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Teletype
dpkg-preconfigure: unable to re-open stdin:
Selecting previously unselected package zstd.
(Reading database ... 127332 files and directories currently installed.)
Preparing to unpack .../zstd_1.5.5+dfsg2-2build1.1_amd64.deb ...
Unpacking zstd (1.5.5+dfsg2-2build1.1) ...
Setting up zstd (1.5.5+dfsg2-2build1.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
>>> Installing ollama to /usr/local
>>> Downloading ollama-linux-amd64.tar.zst
######################################################################## 100.0%
>>> Creating ollama user...
>>> Adding ollama user to video group...
>>> Adding current user to ollama group...
>>> Creating ollama systemd service...
WARNING: systemd is not running
WARNING: Unable to detect NVIDIA/AMD GPU. Install lspci or lshw to automatically detect and install GPU dependencies.
>>> The Ollama API is now available at 127.0.0.1:11434.
>>> Install complete. Run "ollama" from the command line.




NAME                        ID              SIZE      MODIFIED
qwen3:8b                    500a1f067a9f    5.2 GB    Less than a second ago
phi4-mini:latest            78fad5d182a7    2.5 GB    55 seconds ago
gemma3:12b                  b92b218582c5    8.2 GB    About a minute ago
gemma3:4b                   68ceab7e468a    3.4 GB    2 minutes ago
mxbai-embed-large:latest    468836162de7    669 MB    3 minutes ago
```

C# Setup

```
!wget -q https://dot.net/v1/dotnet-install.sh
!bash dotnet-install.sh --channel 10.0 --install-dir /content/dotnet
```

```
import os
os.environ["PATH"] = "/content/dotnet:" + os.environ["PATH"]
```

```
dotnet-install: Attempting to download using aka.ms link https://builds.dotnet.microsoft.com/dotnet/Sdk/10.0.402/dotnet-sdk-10.0.402-linux-x64.tar.gz
dotnet-install: Remote file https://builds.dotnet.microsoft.com/dotnet/Sdk/10.0.402/dotnet-sdk-10.0.402-linux-x64.tar.gz size is 240279509 bytes.
dotnet-install: Extracting archive from https://builds.dotnet.microsoft.com/dotnet/Sdk/10.0.402/dotnet-sdk-10.0.402-linux-x64.tar.gz
dotnet-install: Downloaded file size is 240279509 bytes.
dotnet-install: The remote and local file sizes are equal.
dotnet-install: Installed version is 10.0.402
dotnet-install: Adding to current process PATH: `/content/dotnet`. Note: This change will be visible only when sourcing script.
dotnet-install: Note that the script does not resolve dependencies during installation.
dotnet-install: To check the list of dependencies, go to https://learn.microsoft.com/dotnet/core/install, select your operating system and check the "Dep
dotnet-install: Installation finished successfully.
```

Config

```
# Behavior Profiles
test_behavior_profile_AggressiveMelee = True
test_behavior_profile_Doctor = True
test_behavior_profile_Guardian = True

# Scenario Chunks
test_scenario_1 = True
test_scenario_2 = True
test_scenario_3 = True
```

Behavior Profile

```
!dotnet new console -o BehaviorGenerator
!dotnet new console -o GameSimulator
```

```
□=□=□=□=□=□=□=□=□=□=
Welcome to .NET 10.0!
---------------------
SDK Version: 10.0.402

Telemetry
---------
The .NET tools collect usage data in order to help us improve your experience. It is collected by Microsoft and shared with the community. You can opt-out

Read more about .NET CLI Tools telemetry: https://aka.ms/dotnet-cli-telemetry

----------------
Installed an ASP.NET Core HTTPS development certificate.
To trust the certificate, run 'dotnet dev-certs https --trust'
Learn about HTTPS: https://aka.ms/dotnet-https
```

```
----------------
Write your first app: https://aka.ms/dotnet-hello-world
Find out what's new: https://aka.ms/dotnet-whats-new
Explore documentation: https://aka.ms/dotnet-docs
Report issues and find source on GitHub: https://github.com/dotnet/core
Use 'dotnet --help' to see available commands or visit: https://aka.ms/dotnet-cli
--------------------------------------------------------------------------------
The template "Console App" was created successfully.

Processing post-creation actions...
Restoring /content/BehaviorGenerator/BehaviorGenerator.csproj:
  Determining projects to restore...
  Restored /content/BehaviorGenerator/BehaviorGenerator.csproj (in 121 ms).
Restore succeeded.


□=□=□=The template "Console App" was created successfully.

Processing post-creation actions...
Restoring /content/GameSimulator/GameSimulator.csproj:
  Determining projects to restore...
  Restored /content/GameSimulator/GameSimulator.csproj (in 101 ms).
Restore succeeded.
```

```
%%writefile BehaviorGenerator/behavior_profles.cs
using System;
using System.IO;
using System.Collections.Generic;
using System.Text.Json;

public class BehaviorProfile
{
    public string Name { get; set; }
    public string Description { get; set; }
    public List<string> ScenarioChunks { get; set; }

    public BehaviorProfile(string name, string description)
    {
        Name = name;
        Description = description;
        ScenarioChunks = new List<string>();
    }

    public string ToJson()
    {
        return JsonSerializer.Serialize(this, new JsonSerializerOptions { WriteIndented = true });
    }
}

public class Program
```

```
{
    public static void Main()
    {
        var behaviorProfiles = new List<BehaviorProfile>();

        // Building Aggressive_Melee Behavior Profile
        string aggressive_melee_description = "An aggressive agent that prefers to attack rather than defend itself or others. The agent is only capable
        var Aggressive_Melee = new BehaviorProfile("Aggressive Melee", aggressive_melee_description);

        Aggressive_Melee.ScenarioChunks.Add("When enemy is low HP, the agent will prioritize that target.");
        Aggressive_Melee.ScenarioChunks.Add("Agent does not attack allies.");
        Aggressive_Melee.ScenarioChunks.Add("Agent tend to target the closest enemy");
        Aggressive_Melee.ScenarioChunks.Add("When there are no enemies, agent will return to starting location");
        Aggressive_Melee.ScenarioChunks.Add("If Enemies are out of range the agent will need to approach its target");

        behaviorProfiles.Add(Aggressive_Melee);

        // Building Doctor Behavior Profile
        string doctor_description = "An agent without any combat capabilities. Helps allies by healing and supporting them.";
        var Doctor = new BehaviorProfile("Doctor", doctor_description);

        Doctor.ScenarioChunks.Add("When an ally has low HP, the agent will prioritize healing that ally.");
        Doctor.ScenarioChunks.Add("When multiple allies are injured, the agent will prioritize the ally with the lowest HP percentage.");
        Doctor.ScenarioChunks.Add("Agent cannot attack enemies under any circumstances.");
        Doctor.ScenarioChunks.Add("When an enemy approaches, the agent will attempt to retreat to a safe location.");
        Doctor.ScenarioChunks.Add("When no allies require healing, the agent will remain near friendly units.");
        Doctor.ScenarioChunks.Add("If an injured ally is out of healing range, the agent will approach that ally.");
        Doctor.ScenarioChunks.Add("Agent will prioritize its own survival when threatened by nearby enemies.");

        behaviorProfiles.Add(Doctor);

        // Building Guardian Behavior Profile
        string guardian_description = "An agent with high defensive capabilties. Protects allies and prioritizes survival. Capable of attacking but will
        var Guardian = new BehaviorProfile("Guardian", guardian_description);

        Guardian.ScenarioChunks.Add("When an ally is threatened by an enemy, the agent will prioritize protecting that ally.");
        Guardian.ScenarioChunks.Add("When multiple allies are threatened, the agent will prioritize protecting the ally with the lowest HP.");
        Guardian.ScenarioChunks.Add("Agent will prioritize defensive actions over offensive actions.");
        Guardian.ScenarioChunks.Add("When an enemy approaches a vulnerable ally, the agent will attempt to intercept that enemy.");
        Guardian.ScenarioChunks.Add("Agent will attack enemies when doing so does not compromise the safety of its allies.");
        Guardian.ScenarioChunks.Add("When the agent has low HP, it will prioritize defensive actions and survival.");
        Guardian.ScenarioChunks.Add("When no allies are threatened, the agent will remain near friendly units.");
        Guardian.ScenarioChunks.Add("Agent will avoid pursuing distant enemies if doing so leaves allies unprotected.");

        behaviorProfiles.Add(Guardian);

        // Converting to json
        foreach (var profile in behaviorProfiles)
        {
            // Converting the profile list to JSON
            string json = JsonSerializer.Serialize(profile, new JsonSerializerOptions { WriteIndented = true });
```

```
            // Saving JSON to a file
            File.WriteAllText("behavior_profile_" + profile.Name + ".json", json);
        }

    }
}
```

```
Writing BehaviorGenerator/behavior_profles.cs
```

Game Context

```
%%writefile GameSimulator/game_context.cs
using System;
using System.IO;
using System.Collections.Generic;
using System.Text.Json;


public class GameContext
{
    public List<string> ValidTransitionStates{ get; set; }
    public List<string> Context { get; set; }

    public GameContext()
    {
        ValidTransitionStates = new List<string>();
        Context  = new List<string>();
    }

    public string ToJson()
    {
        return JsonSerializer.Serialize(this, new JsonSerializerOptions { WriteIndented = true });
    }
}

public class Character
{
    //TBD
}

public class State
{
    //TBD
}
```

```
public class Program
{
    public static void Main()
    {
        var GameContexts = new List<GameContext>();

        // Building State List
        var AllStates = new List<string>();
        AllStates.Add("Attack");
        AllStates.Add("Ranged_Attack");
        AllStates.Add("Defend_Self");
        AllStates.Add("Evade");
        AllStates.Add("Approach");
        AllStates.Add("Retreat");
        AllStates.Add("Return");
        AllStates.Add("Heal");
        AllStates.Add("Buff_Ally");
        AllStates.Add("Debuff_Enemy");
        AllStates.Add("Guard_Ally");
        AllStates.Add("Intercept");
        AllStates.Add("Patrol");
        AllStates.Add("Chase");
        AllStates.Add("Take_Cover");
        AllStates.Add("Hold_Position");
        AllStates.Add("Reposition");
        AllStates.Add("Use_Item");
        AllStates.Add("Wait");
        AllStates.Add("Flee");

        // Creating First Game Scenario Context
        var GameContext1 = new GameContext();
        GameContext1.ValidTransitionStates = AllStates;

        GameContext1.Context.Add("Agent Attack Range: 1m");
        GameContext1.Context.Add("Enemy Count: 2");
        GameContext1.Context.Add("Enemy A State: Approach");
        GameContext1.Context.Add("Enemy A Target: Ally A");
        GameContext1.Context.Add("Enemy A Distance: 5m");
        GameContext1.Context.Add("Enemy A HP: 10/20 (50%)");
        GameContext1.Context.Add("Enemy B State: Flee");
        GameContext1.Context.Add("Enemy B Distance: 20m");
        GameContext1.Context.Add("Enemy B HP: 1/20 (5%)");
        GameContext1.Context.Add("Ally A State: Defend_Self");
        GameContext1.Context.Add("Ally A Distance: 5m");
        GameContext1.Context.Add("Ally A HP: 10/20 (50%)");

        GameContexts.Add(GameContext1);


        // Creating Second Game Scenario Context
        var GameContext2 = new GameContext();
```

```
GameContext2.ValidTransitionStates = AllStates;

GameContext2.Context.Add("Agent Attack Range: 1m");
GameContext2.Context.Add("Enemy Count: 2");
GameContext2.Context.Add("Enemy A State: Guard");
GameContext2.Context.Add("Enemy A Distance: 1m");
GameContext2.Context.Add("Enemy A HP: 1/20 (5%)");
GameContext2.Context.Add("Enemy B State: Attack");
GameContext2.Context.Add("Enemy B Target: Agent");
GameContext2.Context.Add("Enemy B Distance: 1m");
GameContext2.Context.Add("Enemy B HP: 20/20 (100%)");

GameContexts.Add(GameContext2);


// Creating Third Game Scenario Context
// Expected: Attack Enemy B
var GameContext3 = new GameContext();
GameContext3.ValidTransitionStates = AllStates;

GameContext3.Context.Add("Agent Attack Range: 1m");
GameContext3.Context.Add("Enemy Count: 2");
GameContext3.Context.Add("Enemy A State: Attack");
GameContext3.Context.Add("Enemy A Target: Agent");
GameContext3.Context.Add("Enemy A Distance: 1m");
GameContext3.Context.Add("Enemy A HP: 20/20 (100%)");
GameContext3.Context.Add("Enemy B State: Flee");
GameContext3.Context.Add("Enemy B Distance: 1m");
GameContext3.Context.Add("Enemy B HP: 1/20 (5%)");

GameContexts.Add(GameContext3);


// Creating Fourth Game Scenario Context
var GameContext4 = new GameContext();
GameContext4.ValidTransitionStates = AllStates;

GameContext4.Context.Add("Agent Attack Range: 1m");
GameContext4.Context.Add("Enemy Count: 0");
GameContext4.Context.Add("Ally Count: 1");
GameContext4.Context.Add("Ally A Staet: Idle");
GameContext4.Context.Add("Ally A Distance: 1m");
GameContext4.Context.Add("Ally A HP: 10/20 (50%)");

GameContexts.Add(GameContext4);


// Saving JSON to a file
int i = 1;
foreach (var gameContext in GameContexts)
{
```

```
            string json = gameContext.ToJson();
            File.WriteAllText("game_context_" + i + ".json", json);
            i++;
        }
    }
}
```

```
Writing GameSimulator/game_context.cs
```

Running C# Scripts

```
# Run C# Scripts
!rm BehaviorGenerator/Program.cs
!dotnet run --project BehaviorGenerator
!rm GameSimulator/Program.cs
!dotnet run --project GameSimulator
```

```
□=□=□=□=
```

Deserialization

```
import json

def Deserialize(json_path):
    # Loading JSON
    with open(json_path, "r") as file:
        deserializedObject = json.load(file)

    return deserializedObject
```

Scenario Chunks Database

```
from langchain_ollama import OllamaEmbeddings
from langchain_community.vectorstores import Chroma

# Setting up langchain vectore store....Should make for every AI Agent using framework?
embeddings = OllamaEmbeddings(model="mxbai-embed-large", keep_alive=7200)

def Build_Collection(collection_name, embedding_function):
    collection = Chroma(
```

```
        collection_name=collection_name,
        embedding_function=embeddings
    )

    return collection
```

```
/tmp/ipykernel_697/4094049768.py:2: DeprecationWarning: `langchain-community` is being sunset and is no longer actively maintained. See https://github.co
  from langchain_community.vectorstores import Chroma
```

```
# Builds a collection from a JSON
def Build_Profile_Collection(behavior_profile):

    # Extracting Json profile
    collection_name = "scenario_" + behavior_profile["Name"].lower().replace(" ", "_")
    scenario_chunks = behavior_profile["ScenarioChunks"]

    # Creating Collection for Behavior Profile
    collection = Build_Collection(collection_name, embeddings)

    # Creating Unique IDs for each scenario chunk
    ids = []
    for i in range(len(scenario_chunks)):
        ids.append(str(i))

    # Creating metadata for each scenario chunk
    metadatas = []
    for i in range(len(scenario_chunks)):
        metadatas.append({
            "id": str(i),
            "profile_name": behavior_profile["Name"]
        })

    # Adding chunks to database
    collection.add_texts(
        texts=scenario_chunks,
        ids=ids,
        metadatas=metadatas
    )

    return collection
```

```
# Building Collections for Each Behavior Profile from JSON

Aggressive_Melee = Deserialize("/content/behavior_profile_Aggressive Melee.json")
Aggressive_Melee_Collection = Build_Profile_Collection(Aggressive_Melee)

Doctor = Deserialize("/content/behavior_profile_Doctor.json")
Doctor_Collection = Build_Profile_Collection(Doctor)
```

```
Guardian = Deserialize("/content/behavior_profile_Guardian.json")
Guardian_Collection = Build_Profile_Collection(Guardian)

print(Aggressive_Melee_Collection._collection.name)
print(Doctor_Collection._collection.name)
print(Guardian_Collection._collection.name)
```

```
/tmp/ipykernel_697/4094049768.py:8: LangChainDeprecationWarning: The class `Chroma` was deprecated in LangChain 0.2.9 and will be removed in 1.0. An updat
  collection = Chroma(
scenario_aggressive_melee
scenario_doctor
scenario_guardian
```

Scenario Chunks Test

```
results = Aggressive_Melee_Collection.get()

# convert to normal
ids = results["ids"]
documents = results["documents"]

print("For Collection ",Aggressive_Melee_Collection._collection.name)
for i in range(len(ids)):
    print("ID:", ids[i])
    print("Chunk:", documents[i], "\n")
```

```
For Collection  scenario_aggressive_melee
ID: 0
Chunk: When enemy is low HP, the agent will prioritize that target.

ID: 1
Chunk: Agent does not attack allies.

ID: 2
Chunk: Agent tend to target the closest enemy

ID: 3
Chunk: When there are no enemies, agent will return to starting location

ID: 4
Chunk: If Enemies are out of range the agent will need to approach its target
```

Game Context

```
Scenario_1 = Deserialize("/content/game_context_1.json")
Scenario_2 = Deserialize("/content/game_context_2.json")
```

```
Scenario_3 = Deserialize("/content/game_context_3.json")
Scenario_4 = Deserialize("/content/game_context_4.json")

Scenarios = [Scenario_1, Scenario_2, Scenario_3, Scenario_4]
```

State Selector Query Method

```
from ollama import Client

# Prepping LLM
alpaca = Client()
```

```
import textwrap

def select_state(scenario, behavior_profile, collection, model, temperature=0.1, debug=False):

    # Converting to single string
    game_context = "\n".join(scenario["Context"])
    valid_states = "\n".join(scenario["ValidTransitionStates"])


    # Retrieval
    retrieved_chunks = collection.similarity_search_with_relevance_scores(
        game_context,
        k=3,
        score_threshold=0.4
    )

    retrieved_scenario_chunks = ""

    for document, score in retrieved_chunks:
        retrieved_scenario_chunks += document.page_content + "\n"

        if debug:
            print("ID:", document.metadata["id"])
            print("Chunk:", document.page_content)
            print("Score:", score)

    # Prompt
    prompt = (
    "Behavior profile of the Agent:\n"
    f"{behavior_profile["Description"]}\n\n"
    "Current runtime game context:\n"
    f"{game_context}\n\n"
    "Scenario based behavioral context:\n"
    f"{retrieved_scenario_chunks}\n"
    "With the above context select the best state for the agent to transition to."
```

```
        "Only select a state from the below valid states."
        "Select only ONE of the valid states."
        "Do not select state/action that is impossible for the agent to perform."
        "Reponse should be limited to only the selected state and the target, nothing else."
        "If there are no valid states repond with INVALID"
        "Valid States:\n"
        f"{valid_states}"
        "\n"
        "Follow this response format:\n"
        "STATE: <state>\n"
        "TARGET: <target>\n"
        "If the selected state does not require a target respond with this:\n"
        "STATE: <state>\n"
        "TARGET: NONE\n"
        "If there are no valid states repond with:\n"
        "STATE: INVALID\n"
        "TARGET: NONE\n"
        )

        if debug:
            print("_________________\n")
            print("Prompt:")
            print(prompt, "\n")

        # LLM
        response = alpaca.chat(
            model=model,
            messages=[ {"role": "user", "content": prompt}],
            options={ "temperature": temperature},
        )

        return response
```

MODEL DIRTY TESTS

```
!ollama ps
```

```
NAME                        ID              SIZE      PROCESSOR    CONTEXT    RUNNER      UNTIL
mxbai-embed-large:latest    468836162de7    624 MB    100% GPU     512        llamacpp    2 hours from now
```

```
def Query_Scenario_List(scenario_list, behavior_profile, collection, model):
    i = 1
    for scenario in scenario_list:
        response = select_state(
            scenario,
            behavior_profile,
```

```
            collection,
            model=model,
            temperature=0.1,
            debug=False
        )

        print("Behavior: ", behavior_profile["Name"])
        print("Response to Scenario ", i, ":")
        print(response["message"]["content"], "\n")
        i = i + 1
```

phi4-mini

```
# Scenario 1, Including Prompt Printing
response = select_state(
    Scenario_1,
    Aggressive_Melee,
    Aggressive_Melee_Collection,
    model="phi4-mini",
    temperature=0.1,
    debug=True
)

print("Response to Scenario 1:")
print(response["message"]["content"], "\n")


Query_Scenario_List(Scenarios, Aggressive_Melee, Aggressive_Melee_Collection, "phi4-mini")
Query_Scenario_List(Scenarios, Doctor, Doctor_Collection, "phi4-mini")
Query_Scenario_List(Scenarios, Guardian, Guardian_Collection, "phi4-mini")
```

```
TARGET: NONE

Behavior:  Doctor
Response to Scenario  2 :
STATE: Flee
TARGET: NONE

Behavior:  Doctor
Response to Scenario  3 :
STATE: Flee
TARGET: NONE

Behavior:  Doctor
Response to Scenario  4 :
STATE: Heal
TARGET: Ally A

Behavior:  Guardian
Response to Scenario  1 :
STATE: Guard_Ally
TARGET: Ally A

Behavior:  Guardian
Response to Scenario  2 :
STATE: Guard_Ally
TARGET: None

Behavior:  Guardian
Response to Scenario  3 :
STATE: Defend_Self
TARGET: NONE

Behavior:  Guardian
Response to Scenario  4 :
STATE: Guard_Ally
TARGET: Ally A
```

```
!ollama stop phi4-mini
```

gemma3:4b

```
Query_Scenario_List(Scenarios, Aggressive_Melee, Aggressive_Melee_Collection, "gemma3:4b")
Query_Scenario_List(Scenarios, Doctor, Doctor_Collection, "gemma3:4b")
Query_Scenario_List(Scenarios, Guardian, Guardian_Collection, "gemma3:4b")
```

```
Behavior:  Aggressive Melee
Response to Scenario  1 :
STATE: Attack
TARGET: Enemy A
```

```
Behavior:  Aggressive Melee
Response to Scenario  2 :
STATE: Attack
TARGET: Enemy B

Behavior:  Aggressive Melee
Response to Scenario  3 :
STATE: Attack
TARGET: Enemy A

Behavior:  Aggressive Melee
Response to Scenario  4 :
STATE: Attack
TARGET: Ally A

Behavior:  Doctor
Response to Scenario  1 :
STATE: Retreat
TARGET: NONE

Behavior:  Doctor
Response to Scenario  2 :
STATE: Retreat
TARGET: NONE

Behavior:  Doctor
Response to Scenario  3 :
STATE: Retreat
TARGET: NONE

Behavior:  Doctor
Response to Scenario  4 :
STATE: Heal
TARGET: Ally A

Behavior:  Guardian
Response to Scenario  1 :
STATE: Intercept
TARGET: Enemy A

Behavior:  Guardian
Response to Scenario  2 :
STATE: Intercept
TARGET: Enemy B

Behavior:  Guardian
Response to Scenario  3 :
STATE: Intercept
TARGET: Enemy A

Behavior:  Guardian
Response to Scenario  4 :
STATE: Intercept
```

```
!ollama stop gemma3:4b
```

```
Query_Scenario_List(Scenarios, Aggressive_Melee, Aggressive_Melee_Collection, "gemma3:12b")
Query_Scenario_List(Scenarios, Doctor, Doctor_Collection, "gemma3:12b")
Query_Scenario_List(Scenarios, Guardian, Guardian_Collection, "gemma3:12b")
```

```
STATE: Attack
TARGET: Enemy A

Behavior:  Aggressive Melee
Response to Scenario  2 :
STATE: Attack
TARGET: Enemy A

Behavior:  Aggressive Melee
Response to Scenario  3 :
STATE: Attack
TARGET: Enemy A

Behavior:  Aggressive Melee
Response to Scenario  4 :
STATE: Attack
TARGET: Ally A

Behavior:  Doctor
Response to Scenario  1 :
STATE: Heal
TARGET: Ally A

Behavior:  Doctor
Response to Scenario  2 :
STATE: Flee
TARGET: NONE

Behavior:  Doctor
Response to Scenario  3 :
STATE: Retreat
TARGET: NONE

Behavior:  Doctor
Response to Scenario  4 :
STATE: Heal
TARGET: Ally A

Behavior:  Guardian
Response to Scenario  1 :
STATE: Intercept
TARGET: Enemy A

Behavior:  Guardian
Response to Scenario  2 :
STATE: Guard_Ally
TARGET: NONE
```

```
Response to Scenario  3 :
STATE: Guard_Ally
TARGET: NONE

Behavior:  Guardian
Response to Scenario  4 :
STATE: Guard_Ally
TARGET: Ally A
```

```
!ollama stop gemma3:12b
```

```
Query_Scenario_List(Scenarios, Aggressive_Melee, Aggressive_Melee_Collection, "qwen3:8b")
Query_Scenario_List(Scenarios, Doctor, Doctor_Collection, "qwen3:8b")
Query_Scenario_List(Scenarios, Guardian, Guardian_Collection, "qwen3:8b")
```

```
Behavior:  Aggressive Melee
Response to Scenario  1 :
STATE: Approach
TARGET: Enemy A

Behavior:  Aggressive Melee
Response to Scenario  2 :
STATE: Attack
TARGET: Enemy A

Behavior:  Aggressive Melee
Response to Scenario  3 :
STATE: Attack
TARGET: Enemy B

Behavior:  Aggressive Melee
Response to Scenario  4 :
STATE: INVALID
TARGET: NONE

Behavior:  Doctor
Response to Scenario  1 :
STATE: Heal
TARGET: Ally A

Behavior:  Doctor
Response to Scenario  2 :
STATE: Retreat
TARGET: NONE

Behavior:  Doctor
Response to Scenario  3 :
STATE: Retreat
TARGET: NONE

Behavior:  Doctor
Response to Scenario  4 :
```

```
STATE: Heal
TARGET: Ally A

Behavior:  Guardian
Response to Scenario  1 :
STATE: Intercept
TARGET: Enemy A

Behavior:  Guardian
Response to Scenario  2 :
STATE: Defend_Self
TARGET: Enemy B

Behavior:  Guardian
Response to Scenario  3 :
STATE: Defend_Self
TARGET: Enemy A

Behavior:  Guardian
Response to Scenario  4 :
```

```
!ollama stop qwen3:8b
```

```
!ollama ps
```

```
NAME                         ID              SIZE      PROCESSOR    CONTEXT    RUNNER      UNTIL
mxbai-embed-large:latest     468836162de7    624 MB    100% GPU     512        llamacpp    2 hours from now
```